Sarthak Gupta

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Education

University College London — London, UK
Bachelor of Science (Hons.) in Computer Science — *Expected 2029*

Delhi Public School — Gurgaon, India
Central Board of Secondary Education (CBSE) — *August 2016 – May 2026*

- **Standardised Tests:** SAT: 1550 | AP Calculus BC: 5 | AP Computer Science A: 5

Experience

Summer Intern — May 2026 – July 2026
NomuPay — *Remote*

- Built an LLM-to-SQL pipeline to convert natural language questions to database queries and return results from production financial data
- Implemented query generation, schema retrieval, validation and execution, which allowed the agent to handle multi-table queries
- Developed a tool-calling workflow which combined database access and programmatic analysis to generate explanations and surface trends from the underlying data

Student Researcher — September 2025 – December 2025
Venra Tech — *Remote*

- Developed a speech-processing pipeline, including audio standardisation, noise reduction, voice activity detection and signal normalisation for Parkinson's detection using ML
- Implemented extraction and analysis of acoustic, spectral and temporal features, including F0, jitter, shimmer, HNR, MFCCs, formant frequencies and speech-rate features
- Developed an ML classification framework for distinguishing Parkinsonian from healthy speech, and evaluated models using accuracy, precision and recall
- Research supervised by Dr Ritu Arora, *Founder, Venra Tech*; formerly *Research Scientist, Texas Advanced Computing Center (UT Austin)*

Student Researcher — June 2023 – August 2024
Neera Hospital — *Lucknow, India*

- Undertook a novel study on the Indian population and collected and processed a dataset of 2,512 Indian patients
- Trained and evaluated multiple classification models and compared performance using accuracy and other diagnostic metrics
- Obtained 93.3% diagnostic accuracy with Random Forest, which outperformed the other evaluated models; findings were published in the *Journal of Student Research*

Projects

GhostLight: Lighting Estimation for AR | *Python, PyTorch, OpenCV, Blender* — June 2025 – Present

- Built an end-to-end computer vision pipeline that uses deep learning to predict a room's lighting from a single photo, reproducing SIGGRAPH research (Gardner et al., 2017)
- Applied transfer learning to fine-tune a pretrained CNN on a custom dataset of 806 images, generated by data augmentation from 301 open-source HDR indoor panoramas
- Achieved 9.3° lighting-direction error, outperforming the paper's baseline by 6.8%
- In the process of integrating the model into a 3D rendering workflow to realistically light virtual objects in real photos

Reading CEO Uncertainty in Earnings Calls Using AI | *Python, PyTorch* — October 2026 – Present

- Building a multi-agent AI system that gives early warning of stock-price risk by flagging when a CEO becomes unusually evasive on earnings calls, before the market prices it in